

Project XD: SP1

Factors affecting the young generation's intention to engage in smart dairy farm practices in Thailand

การวิเคราะห์ข้อมูลเพื่อหาปัจจัยที่ส่งผลต่อความสนใจในการทำฟาร์มโคนมอัจฉริยะของคนรุ่นใหม่

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Agenda

1

Background & Scope of Work

Background and significant of the study, Objective, Scope of work

2

Related Theories

Theory of Planned Behaviour (TPB), Social Capital Theory, Social Cognitive Theory, Social Learning Theory, Theory Acceptance Model (TAM), Partial Least Squares Structural Equation Modeling (PLS-SEM)

3

Project Planning & Methodology

Project Planning, Methodology; Preliminary data collection and analysis, Factor identification, Questionnaire construction, Pilot test, Analysis

4

Expected Benefits

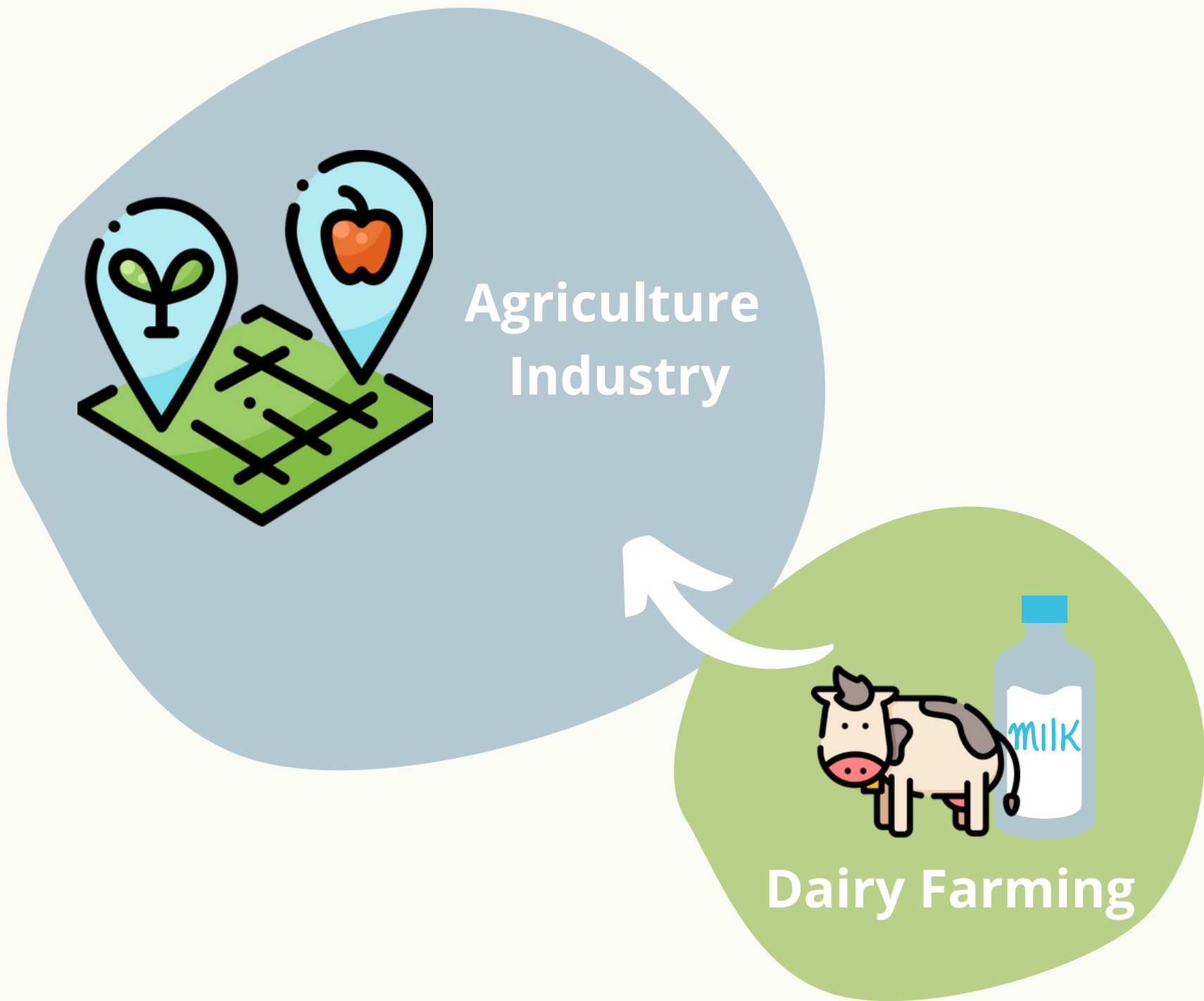
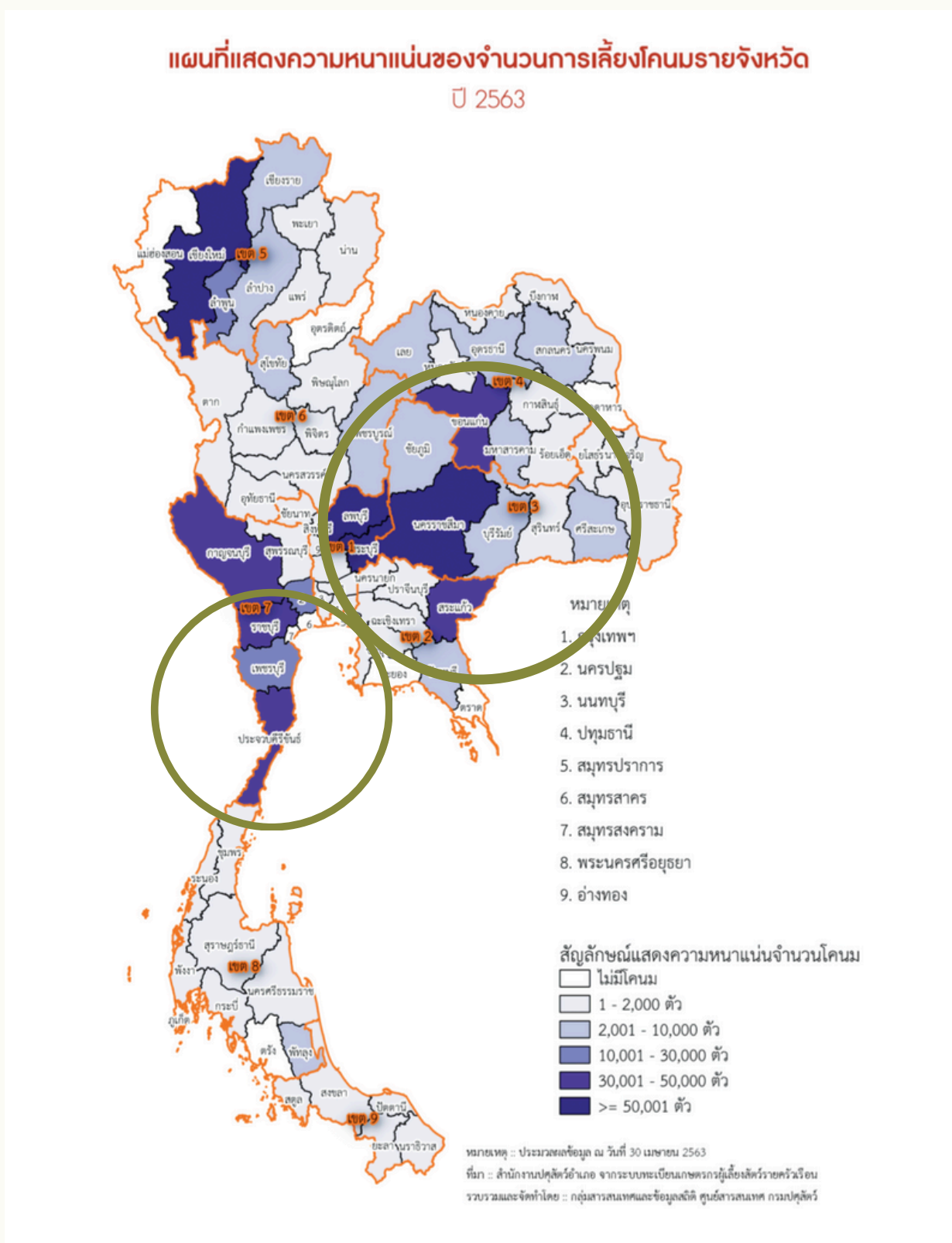
What we planned to achieve

1

Background & Scope of Work

Background and significant of the study, Objective, Scope of work

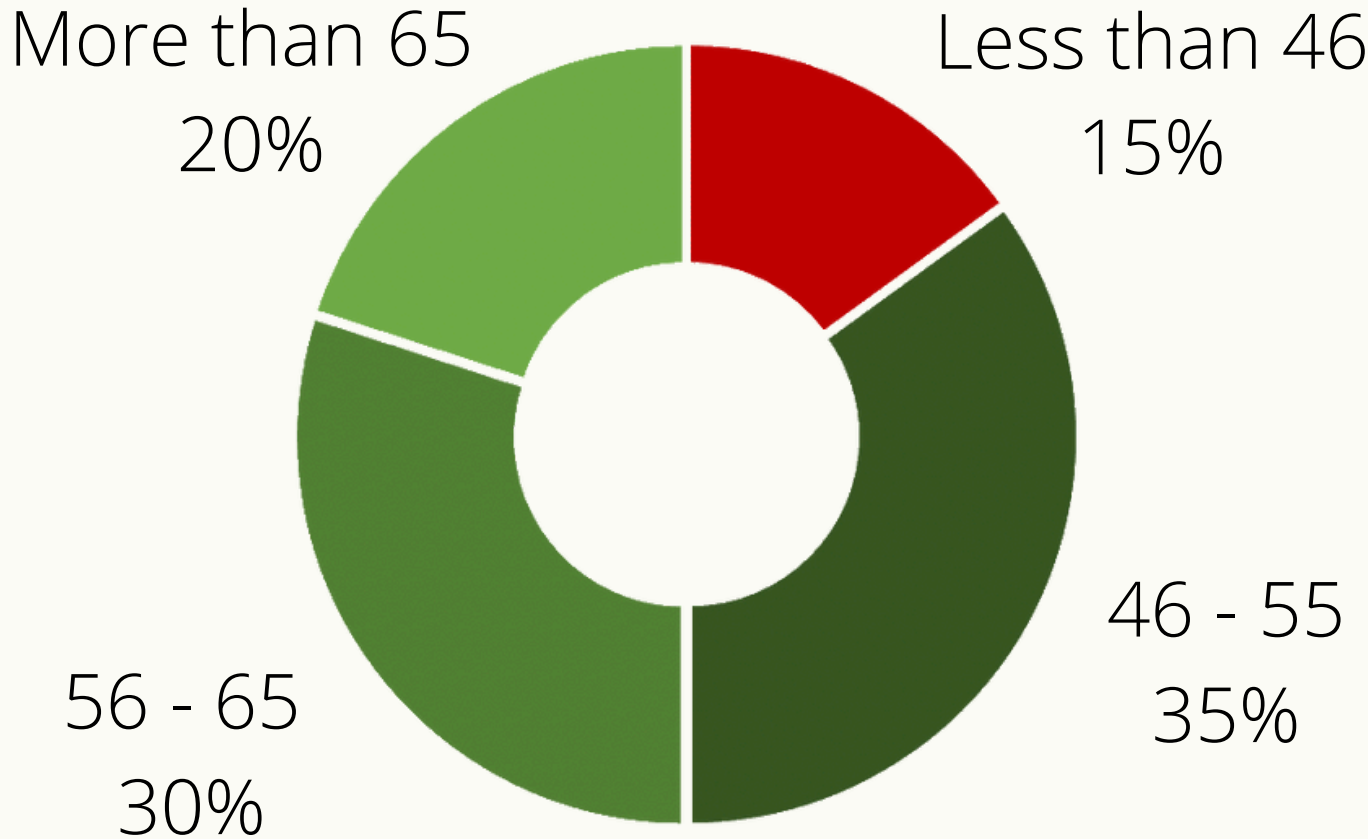
Current Dairy Farming Situation



Reference: ข้อมูลจำนวนปศุสัตว์ในประเทศไทยปี 2562-2563

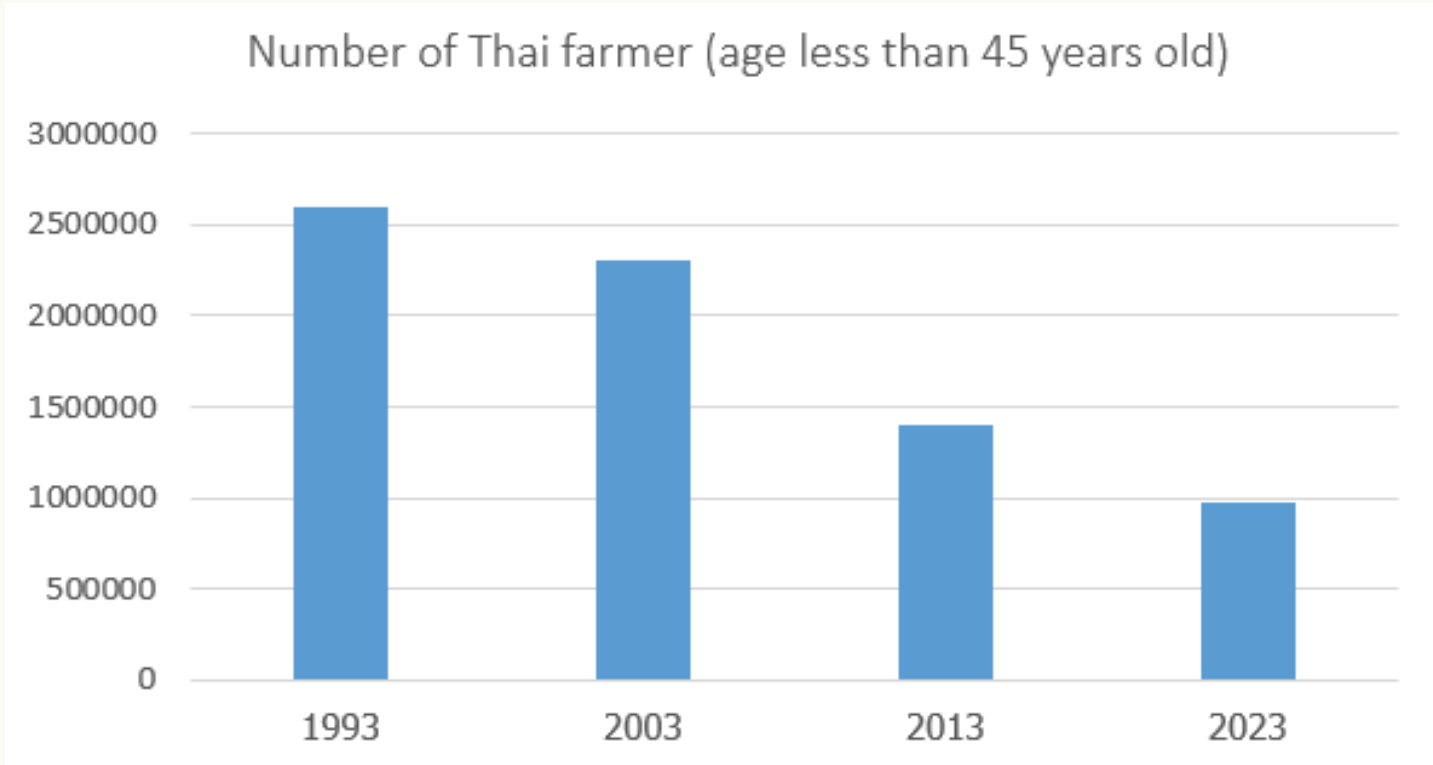
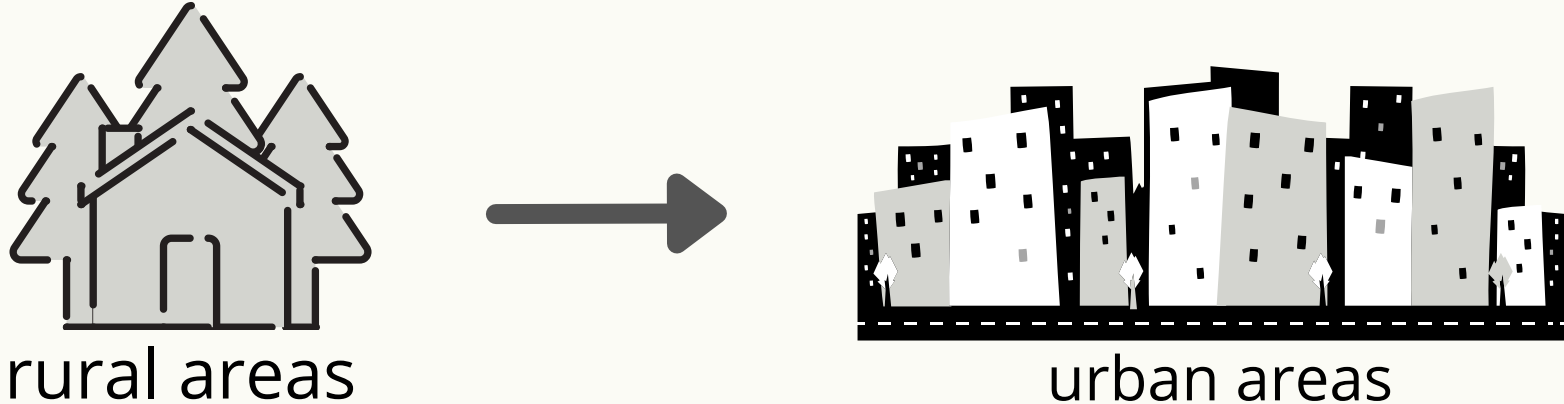
Current Dairy Farming Situation

Percentage of Thai dairy farmers by age



Reference: Kwangsawad and Jattamart, 2020

The rise of urbanization



Reference: Cochetel, Filloux, Salvago, Faysse and W̄yaaŕ, 2020

Current Dairy Farming Situation

General Dairy Farm

Redundant work -> Tedious & Exhausting



Smart Dairy Farm

Report Analysis

Aims: Analyze factors affecting the potential of applying smart technology in the dairy farm business which could be called smart dairy farms

- Focused on the Northeast and West areas of Thailand
- Based on a model of TAM(Theory Acceptance Model), TPB(Theory of Planned Behaviour) and 3 Social Theories

Objective

"To identify factors that influenced the young generation's intention to engage in smart dairy farms in Thailand"

Scope of Work

- Focus on medium-sized dairy farms (21-100 cows) in the Central and North-Eastern of Thailand.
- Focus on the young generation at the age of 20 to 40.
- Identify problems and constraints of dairy farms in Thailand.
- Identify issues that farmers face in engaging in smart dairy farms.
- Conduct questionnaires for in-depth analysis.
- Identify both psychological and social factors that influence farmers to engage in smart dairy farms.

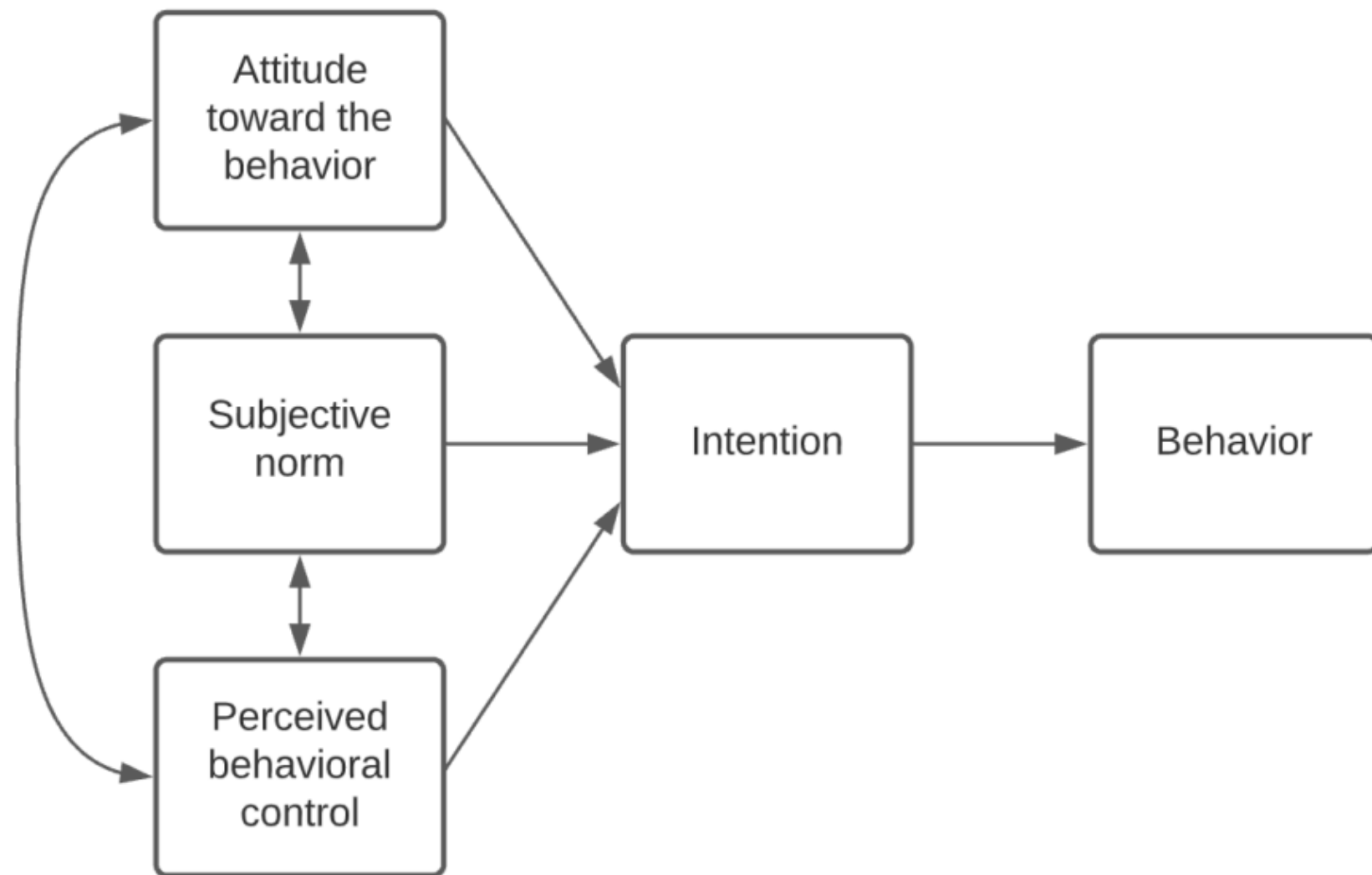
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Related Theories

Theory of Planned Behaviour (TPB), Social Capital Theory, Social Cognitive Theory, Social Learning Theory, Theory Acceptance Model (TAM), Partial Least Squares Structural Equation Modeling (PLS-SEM)

Theory of planned behaviour

The Theory of Planned Behavior (TPB) is a psychological theory that joins beliefs and behavior in order to predict and understand the behavior of individuals (Ajzen, 1991).



Attitude: Attitude refers to the level to which a person has a favorable or unfavorable evaluation of the behavior of interest.

Subjective norm: Subjective norm refers to the belief about whether most people accept or reject such a behavior.

Perceived behavioral control: Perceived behavioral control refers to a person's perception of whether it is easy or difficult to conduct the behavior.

Figure1. Theory of planned behaviour (Adapted from Ajzen, 1991)

Social Capital Theory

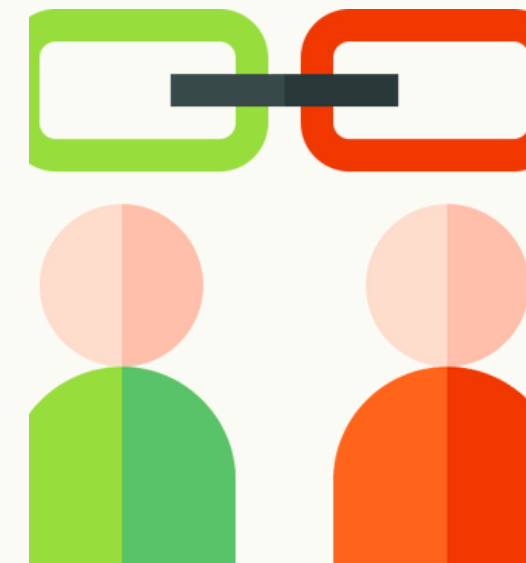
“The aggregate of the actual or potential resources which are linked to possession of a durable network of more or less institutionalized relationships of mutual acquaintance or recognition”



Reputation



Social trust



Connection

Reference : Meijer, 2009, Fukuyama, 1995 and Castillo et al., 2021

Social Cognitive Theory

It has been widely used to understand human's behavioural change that has been influenced by factors including individual experiences, action of others and environment (Bandura, 1977).

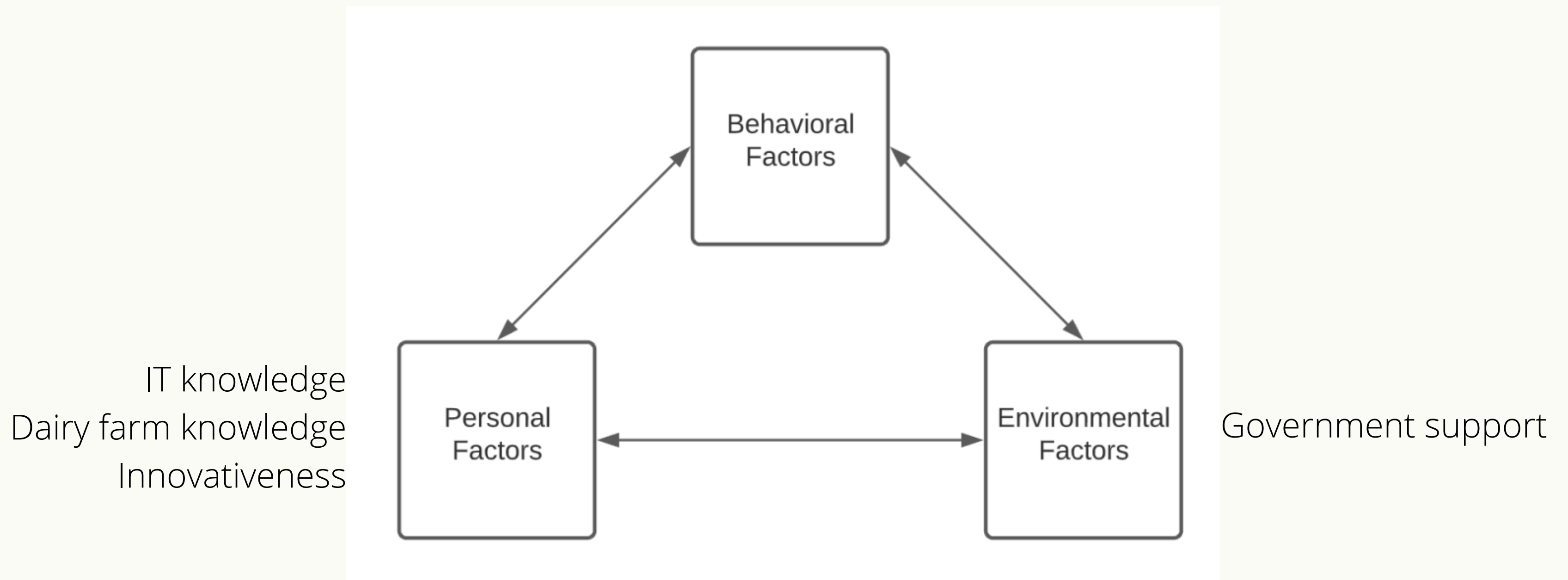


Figure 2. Determinants in Social Cognitive Theory (Adopted from Chin, Mansori, 2018)

Social Learning Theory

Social learning theory suggests that social behavior is learned by observing and imitating the behavior of others (Bandura, 1977)



Figure 3. Social Learning Theory (Adopted from Bandura, 1977)

According to Social Learning theory which suggests that social behavior is learned by observing and imitating the behavior of others,
Role Models can influence the young generation's attitude and subjective norms.

Technology Acceptance Model

TAM model is used to understand how users come to use and accept technology.

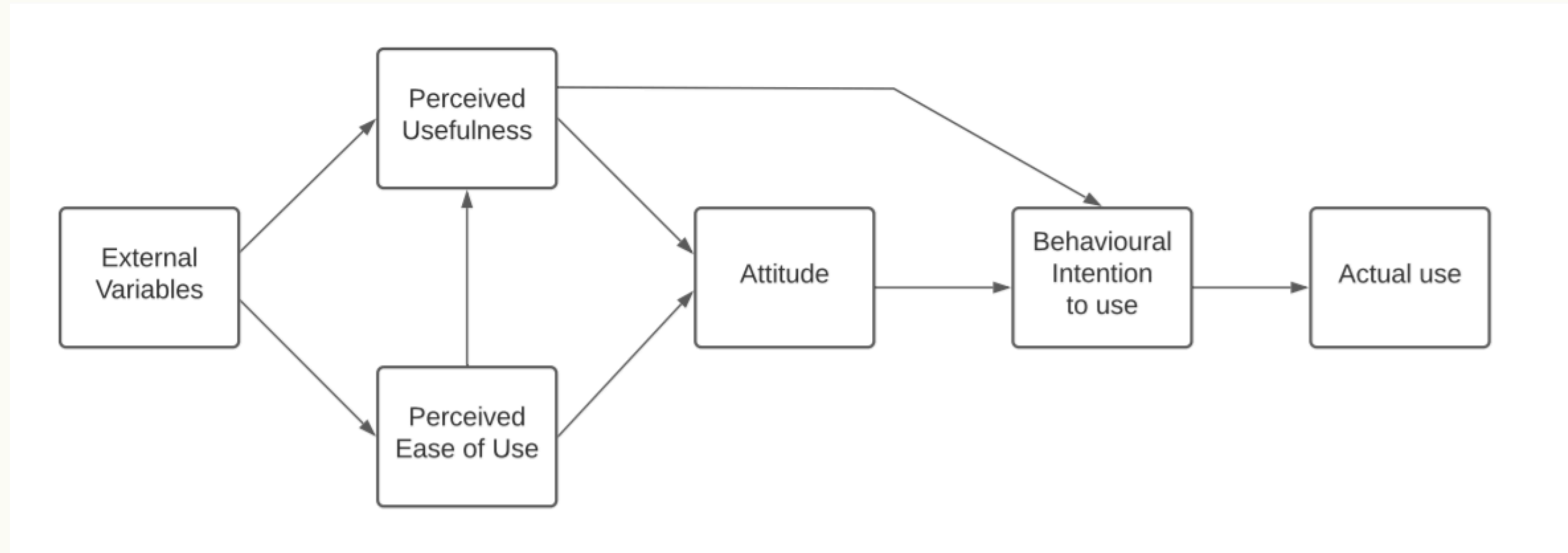


Figure 4. Technology Acceptance Model TAM (Adopted from Davis, 1989)

The perceived usefulness in smart technology is influenced by two external factors including **IT knowledge** and **innovativeness**, which represent the young generation's capability to apply smart technology to dairy farms.

Smart Dairy Farm

Smart farming is a management concept focused on providing the agricultural industry with the infrastructure to leverage advanced technology – including big data, the cloud and the internet of things (IoT)



IoT refers to the internetwork of several nodes integrated with sensors, actuators, electronics equipment and network connectivity to facilitate connectivity, and easy transfer of data in a real-time environment (Mishra, 2020).

A digital bracelet or pedometer



Reference: <https://www.fwi.co.uk/livestock/dairy/next-generation-of-dairy-cow-pedometer-measures-rest-and-activity>

Genomic testing technology



Digitally Signaled Birth



Reference: <https://www.fwi.co.uk/livestock/dairy/next-generation-of-dairy-cow-pedometer-measures-rest-and-activity>

Partial Least Squares Structural Equation Modeling (PLS-SEM)

Partial Least Squares Structural Equation Modeling (PLS-SEM) is usually used to study the relationship between observed and latent variables.

The use of PLS-SEM is for:

1. The sample size is small.
2. There is a shortage of theory for applications.
3. Accuracy in prediction is necessary.
4. The correctness of the model cannot be verified.
5. It can be used to analyse both non-experimental and experimental data.

STEPS

1. Data Preparation
2. Project Creation
3. Building the Inner Model
4. Building the Outer Model
5. Running the Path-Modeling Estimation
6. Assessing the PLS-SEM Output

3

Project Planning & Methodology

Project Planning, Methodology; Preliminary data collection and analysis, Factor identification, Questionnaire construction, Pilot test, Analysis

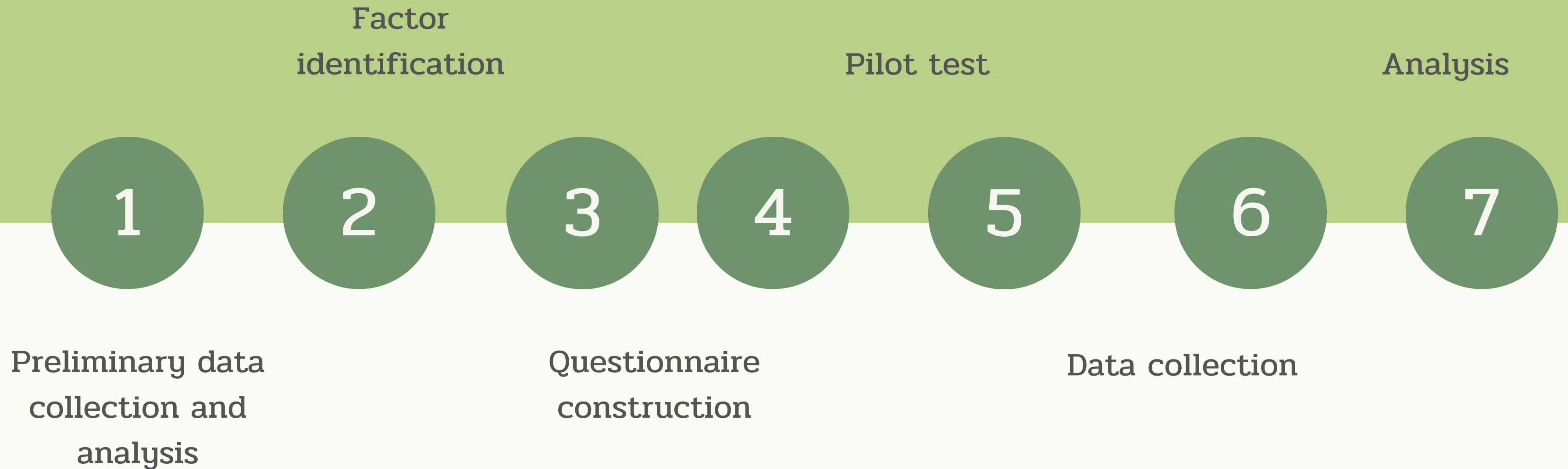
Project Planning

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1	Introduction																																				
	- Background & Motivation																																				
	- Problem Statement & Objective																																				
	- Scope of Work																																				
	- Expected Benefits																																				
2	Literature Review																																				
	- Reseach Model																																				
	- Smart Farm																																				
	- Analysis Tools																																				
3	Preliminary Data Collection and Analysis																																				
4	Factor Identification																																				
5	Questionnaire Construction																																				

Project Planning

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7	Pilot Test																																				
8	Data Collection																																				
9	Data Analysis and Result Discussion																																				
10	Construct Research Report																																				
11	Final Report and Presentation Preparation																																				

Methodology



Preliminary data collection and analysis

- An **expert interview** was conducted to **collect insights and specific information** about the dairy industry. The expert was also asked how to approach dairy farm owners and make a connection with them.
- The **dairy farm supply chain** was studied to understand about the activities involved in processing milk and showed how each stakeholder was linked to each operation.
- **Historical and current statistics** on raw milk production and consumption, and also the number of farmers and dairy cows, were included to help understand **Thailand's position in the dairy industry**.

Factor identification

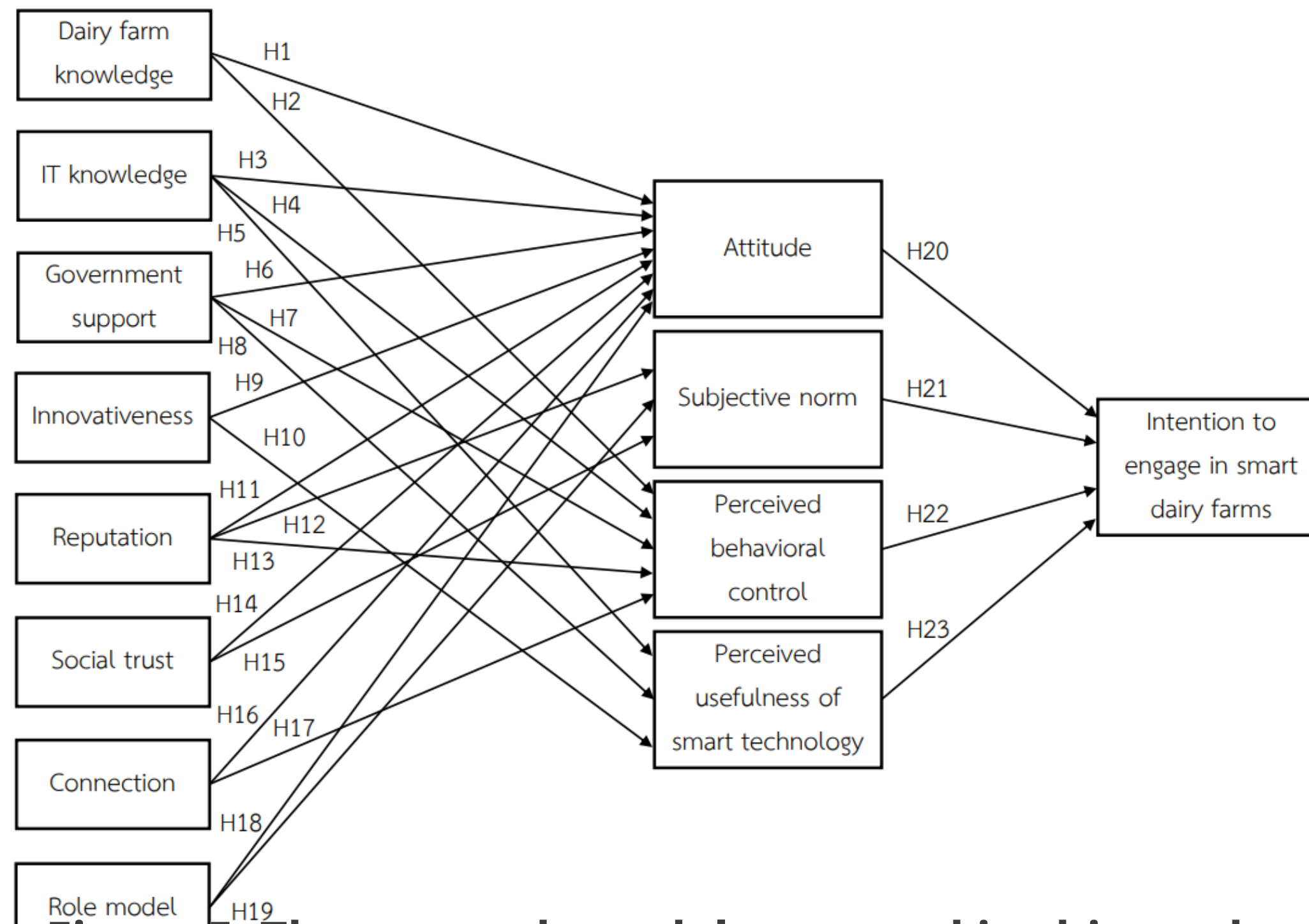


Figure 5. The research model proposed in this study

Factors Identification

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There are 4 factors collected from **Social Cognitive Theory**.

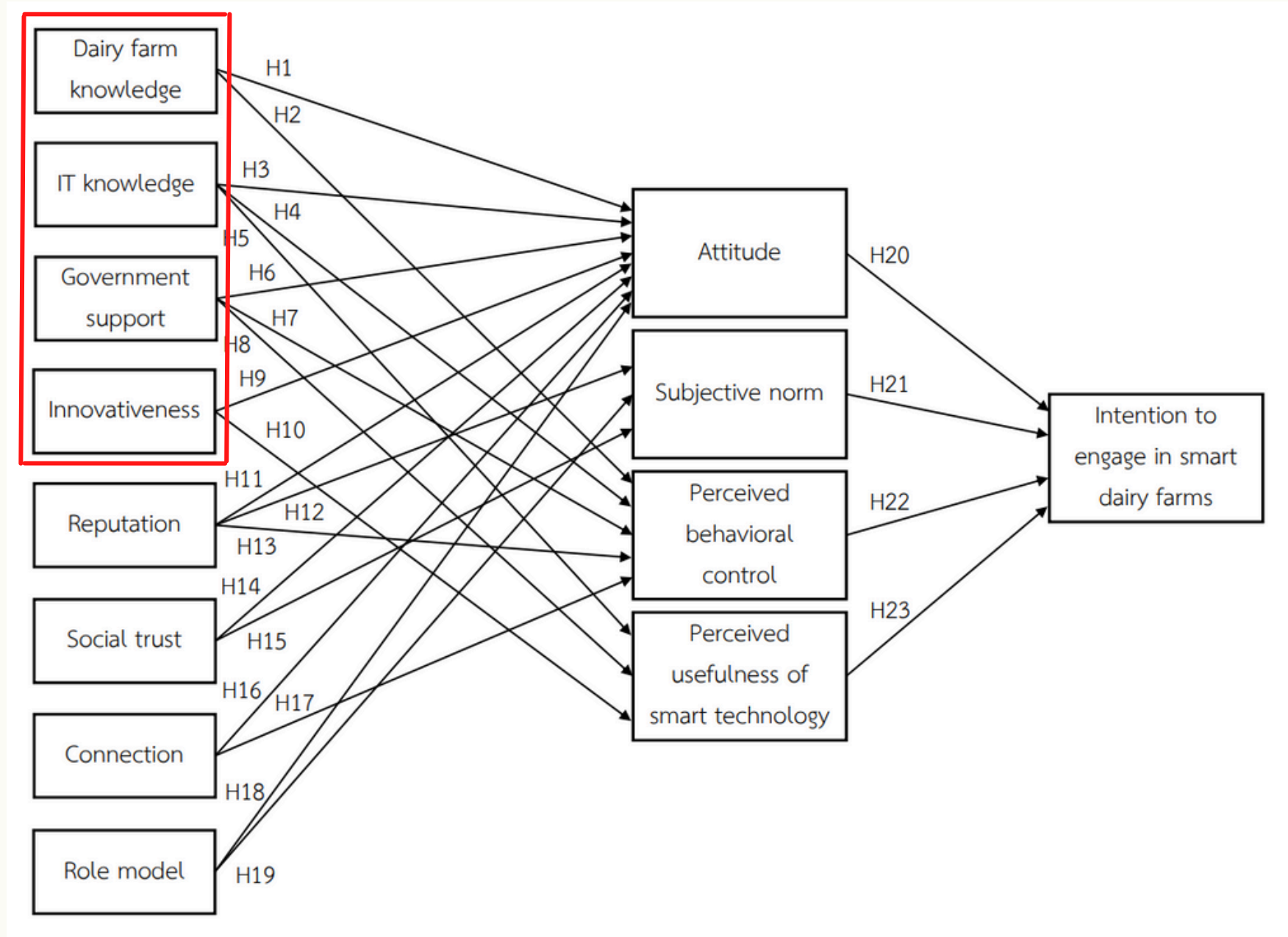


Figure 5. The research model proposed in this study

Factor identification

Social cognitive theory

(1) Dairy farm knowledge: basic knowledge about animals and a good understanding of dairy science to get better quality milk and manage milk production issues.

***H1:** The more understanding of dairy farm knowledge, the more positive the attitude to engage in a smart dairy farms*

***H2:** The more understanding of dairy farm knowledge, the more positive the perceived behavioral control to engage in smart dairy farms*

Factor identification

Social cognitive theory

(2) IT knowledge: ability to learn and understand how to utilize the technologies to make proper management and solve the problems in their farms.

***H3:** The more understanding of IT knowledge, the more positive the attitude to engage in smart dairy farms*

***H4:** The more understanding of IT knowledge, the more positive the perceived behavioral control to engage in smart dairy farms*

***H5:** The more understanding of IT knowledge, the more positive the perceived usefulness of smart technology*

Factor identification

Social cognitive theory

(3) Government support: investment promotion schemes and service providers that enable the digitization of farming by granting a corporate income tax exemption with no annual limit.

***H6:** The higher in government support to promote smart farming, the more positive the attitude to engage in smart dairy farms*

***H7:** The higher in government support to promote smart farming, the more positive the perceived behavioral control to engage in smart dairy farms*

***H8:** The higher in government support to promote smart farming, the more positive the perceived usefulness of smart technology*

Factor identification

Social cognitive theory

(4) Innovativeness: Innovativeness has been described as an internal drive or motivating strength of someone towards novelty and uniqueness.

***H9:** The higher innovativeness of the young generation, the more positive the attitude to engage in smart dairy farms*

***H10:** The higher innovativeness of the young generation, the more positive the perceived usefulness of smart technology*

Factors Identification

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There are 3 factors collected from **Social Capital Theory**.

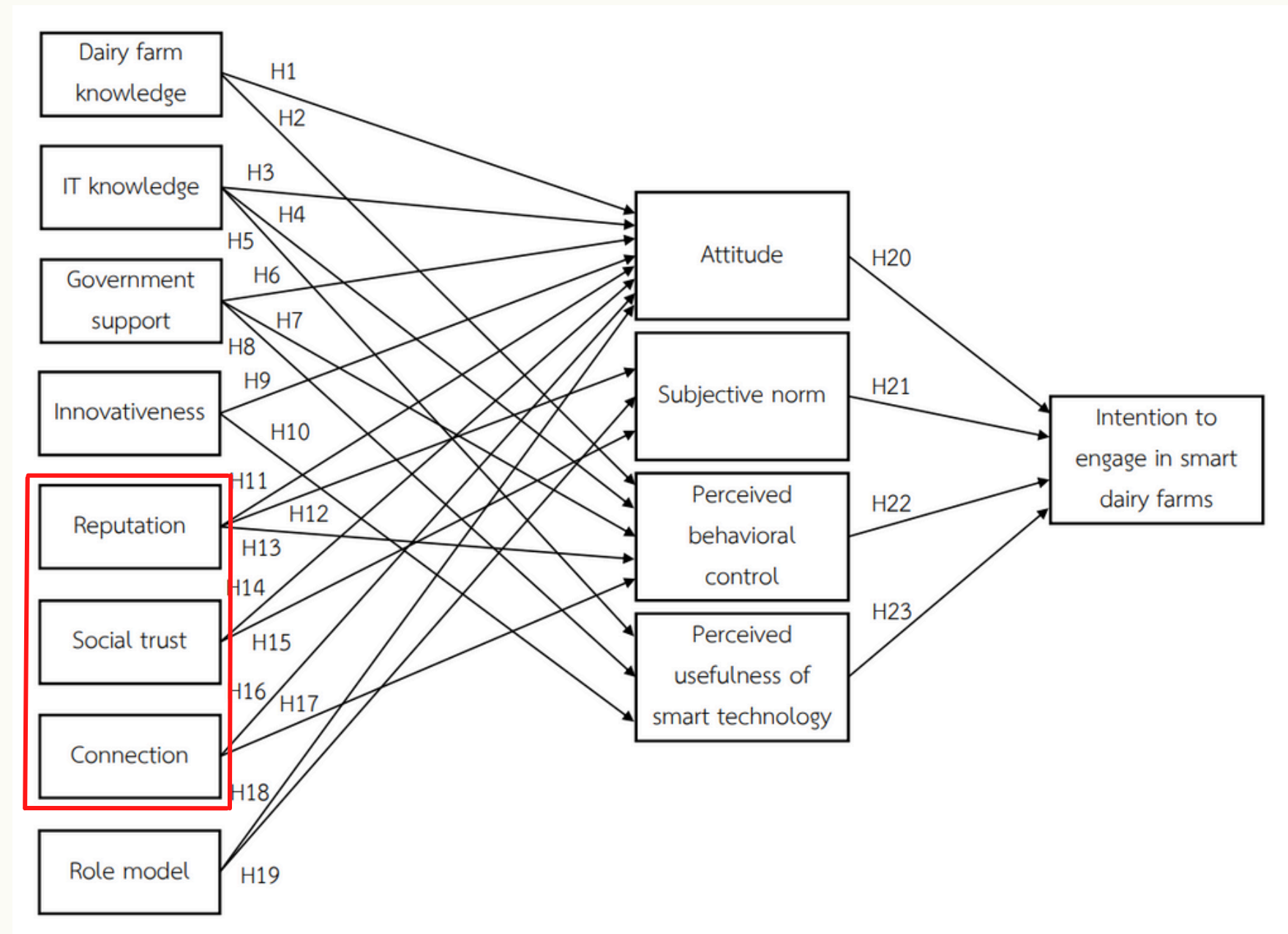


Figure 5. The research model proposed in this study

Factor identification

Social capital theory

(5) Reputation: a critical asset as it may influence an individual's status within a group

H11: The higher the reputation of a dairy farmer, the more positive the attitude to engage in smart dairy farms.

H12: There will be a positive relationship between the reputation of a dairy farmer and the subject norm of the smart dairy farms.

H13: The higher the reputation of a dairy farmer, the more positive the perceived behavioral control to engage in dairy farms

Factor identification

Social capital theory

(6) Social trust: Social trust refers to the expectation about positive future activities that an individual or institution will act competently, fairly, openly, and considerately from past interactions.

***H14:** The higher the social trust among the dairy farmer community, the more positive the attitude to engage in smart dairy farms*

***H15:** There will be a positive relationship between the social trust among the dairy farmer community and the subject norm of the smart dairy farms*

Factor identification

Social capital theory

(7) Connection: Connection refers to the relationships and interactions between members of a collective.

***H16:** The higher the connection among the dairy farmer community, the more positive the attitude to engage in smart dairy farms*

***H17:** The higher the connection among the dairy farmer community, the more positive the perceived behavioral control to engage in smart dairy farm*

Factors Identification

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There is one factors collected from **Social Learning Theory**.

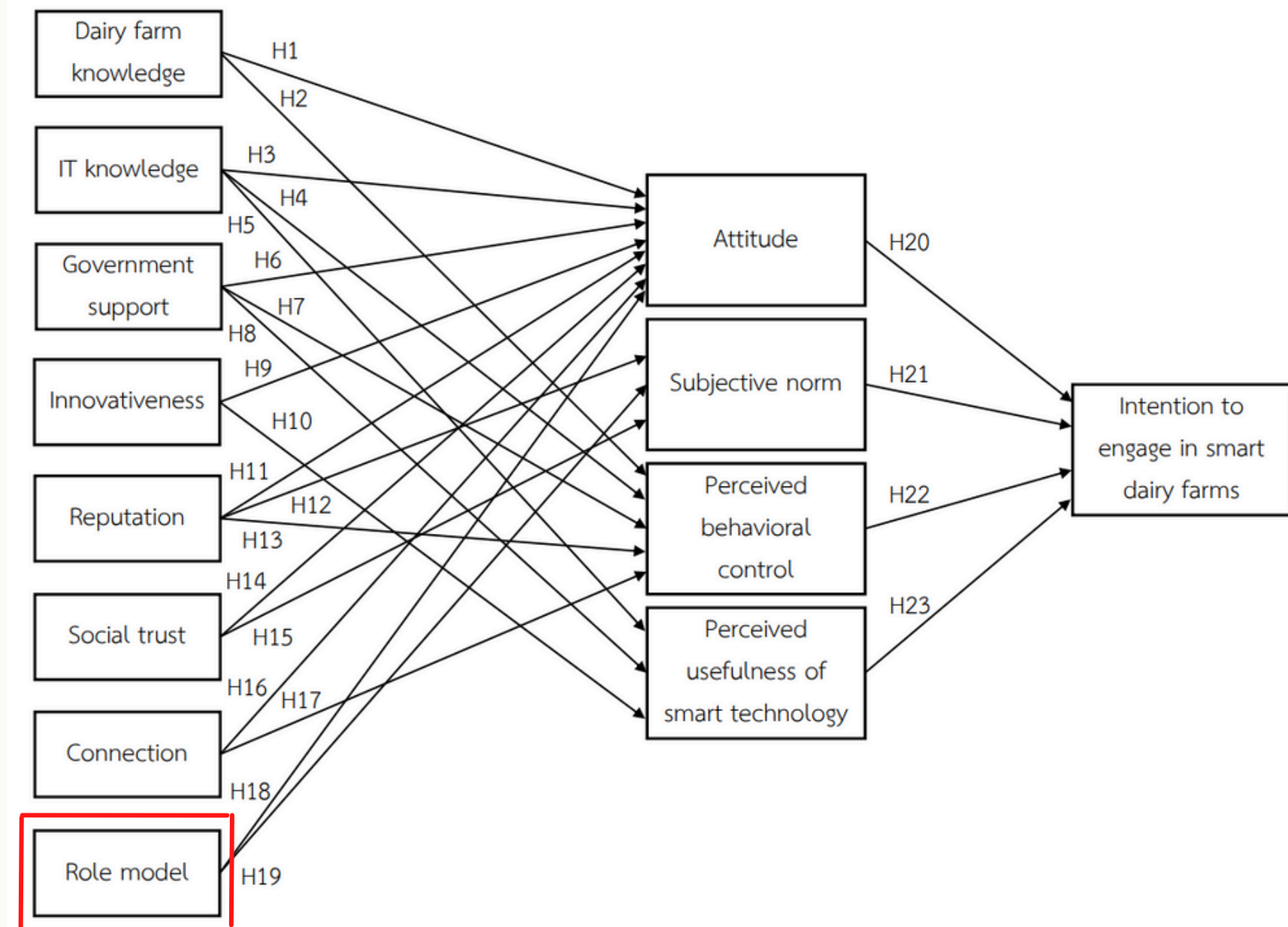


Figure 5. The research model proposed in this study

Factor identification

Social learning theory

(8) Role model: A role model is defined as someone worthy of imitating his or her good behavior.

***H18:** There will be a positive relationship between the role model in dairy farming and the attitude to engage in smart dairy farms.*

***H19:** There will be a positive relationship between the role model in dairy farming and the subjective norm of the smart dairy farms.*

Factors Identification

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The factors in the second layer were collected from **the TPB and TAM model**.

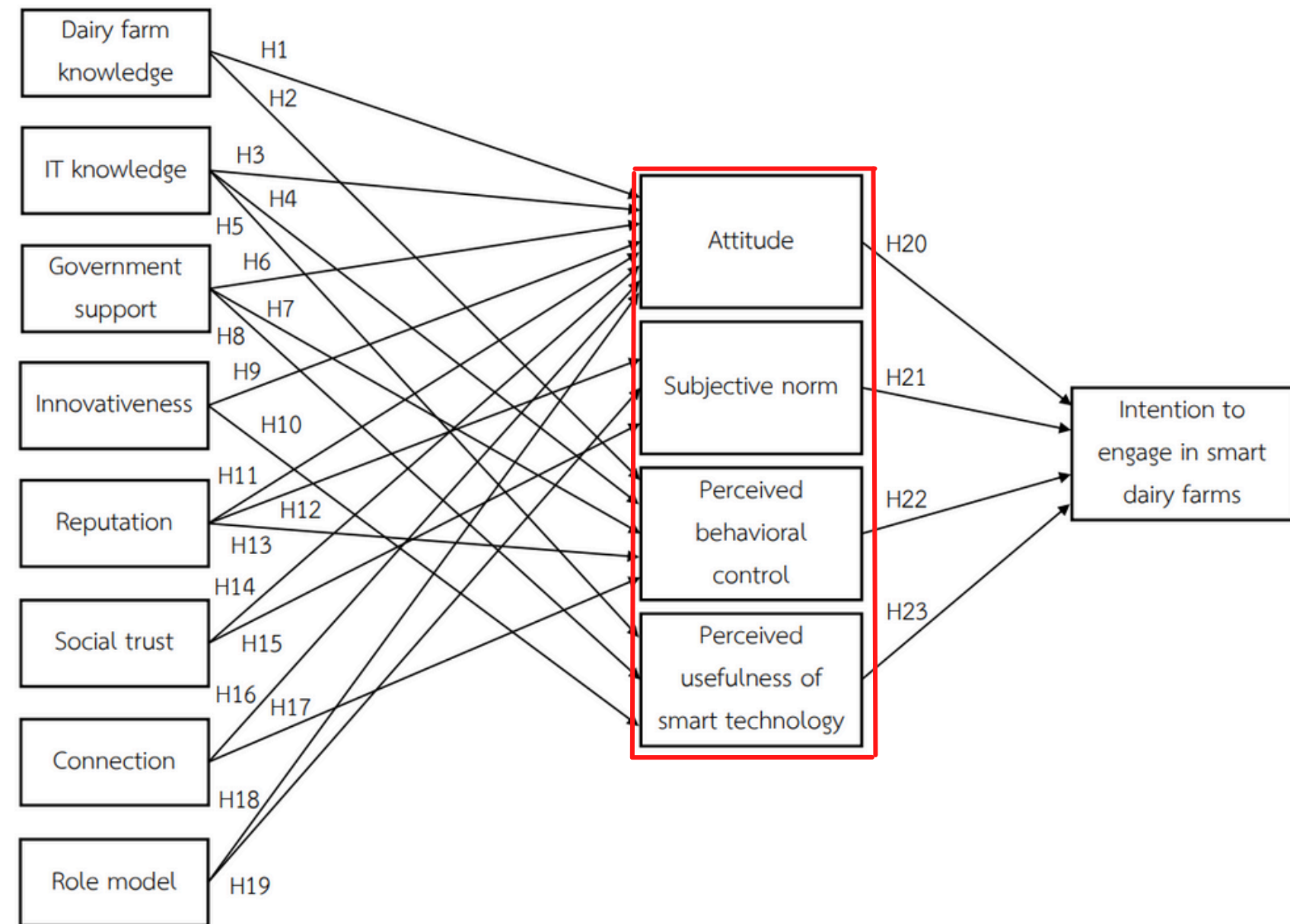


Figure 5. The research model proposed in this study

Factor identification

Theory of Planned Behaviour

(9) Attitude: the level to which a person has a favorable or unfavorable evaluation of the behavior

H20: The young generation's attitude toward smart dairy farming has a positive and significant effect on their intentions to engage in smart dairy farms.

(10) Subjective norm: the belief about whether most people accept or reject such a behavior

H21: The young generation's subjective norm has a positive and significant effect on their intentions to engage in smart dairy farms.

(11) Perceived behavioral control: a person's perception of whether it is easy or difficult to conduct the behavior

H22: The young generation's perceived behavioral control has a positive and significant effect on their intentions to engage in smart dairy farms

Factor identification

Theory Acceptance Model

(12) Perceived usefulness of smart technology: the degree to which a person believes that using smart technology would enhance his/her dairy farming performance

***H23:** The young generation's perceived usefulness of smart technology has a positive and significant effect on their intentions to engage in smart dairy farms*

Factor identification

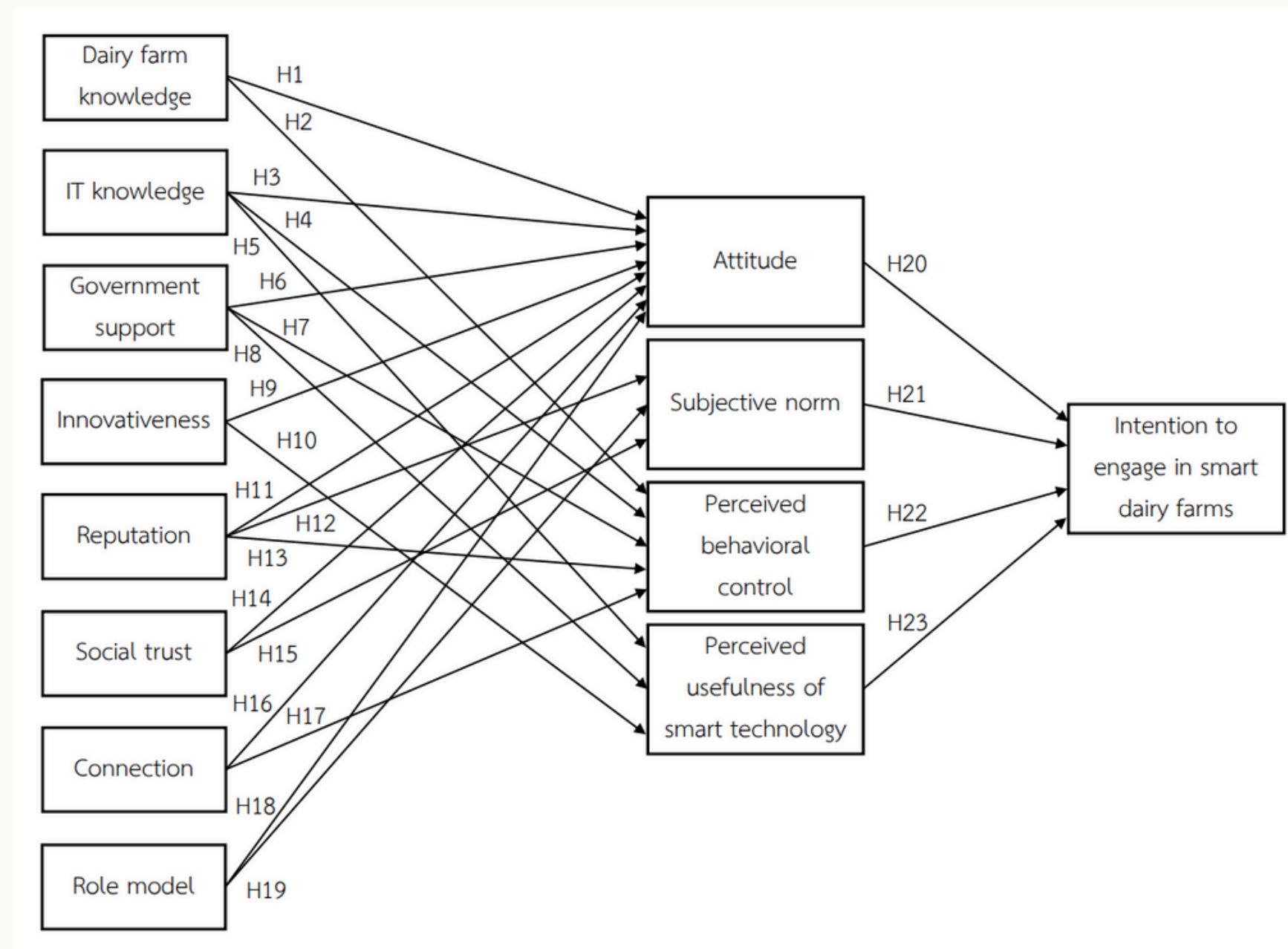


Figure 5. The research model proposed in this study

Questionnaire construction

- The questionnaire is developed by **adapting various questionnaires** from previous studies.
- **The seven-Point Likert Scale** is used to measure the significance of each factor.

Strongly Disagree	Disagree	More or less Disagree	Neutral	More or less agree	Agree	Strongly Agree
1	2	3	4	5	6	7

Ref. Adopted from Rensis Likert (1932)

Pilot test

The questionnaire survey is sent to **20 students majoring in veterinary science**. Then, the stability, consistency and truthfulness of the questionnaire survey are verified by the reliability and validity assessment.

- **Reliability assessment:** the reliability refers to a measurement that supplies consistent results with equal values (Blumberg et al., 2005).
- **Validity assessment:** validity is the extent to which any measuring instrument measures what it is intended to measure (Thatcher, 2010).

Reliability and Validity assessments are conducted by using Smart-PLS software.

Data collection



The data collection of this study is conducted through **online surveys, using a google form**, and sent to the third party to distribute questionnaire surveys.



There are **130 samples from the young generation** (20 - 40 years old) in dairy farm families and the students majoring in agriculture.

Analysis

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The **PLS-SEM** is used to validate the research model. It is also helpful to estimate the significance of relationships between factors that affect the young generation's intention to engage in smart dairy farms. Therefore, **Smart-PLS software** is selected to analyse data collected from questionnaires.

4

Expected Benefits

What we planned to achieve

Expected Benefits

- To be able to **identify the potential problems** of the young generation that impact the engagement in smart dairy farms.
- To provide **guidelines for government and private sectors** to encourage the young generation to engage in smart dairy farms.
- To understand the **young generation's attitude** toward smart dairy farms.

Appendix

Ethical Approval

Attendance No. 2021126

- Attend to Human Subjects Research Ethics Training and obtain an Attendance Number from Tokyo Institute of Technology (Tokyo Tech)

Sample Size

- **10-times rule method** (Hair et al., 2011): The method assumes that the sample size should be greater than ten times of the maximum number of inner or outer model links pointing at any latent variable in the model. There are 13 variables in the model, so a required minimum number of samples is 130 samples.
- **Allowance for error:** The errors may occur while collecting data through questionnaires. As a result, if 20% of the samples are errors during data collection, a minimum number of samples is determined to be 163 samples.

Factor Identification

Title	Researcher	Factor
Towards an Understanding of the Behavioral Intentions and Actual Use of Smart Products among German Farmers	Schukat, S., & Heise, H. (2021)	Social Influence, Technology readiness
Farmers' Knowledge, Attitude, and Adoption of Smart Agriculture Technology in Taiwan	Chuang et al. (2020)	Smart Agriculture (SA) knowledge
A model of farmers intentions towards organic farming: A case study on rice farming in Thailand	Yanakittkul, P., & Aungvaravong, C. (2020)	Support of Government Policy (SGP)
Smallholder farmers' motivations for using Conservation Agriculture and the roles of yield, labour and soil fertility in decision making	Lalani et al. (2016)	Farmers' attitudes toward conservation agriculture, Subjective norm, Perceived behavioral control
Analysis of the ecological conservation behavior of farmers in payment for ecosystem service programs in eco-environmentally fragile areas using social psychology models	Deng et al. (2016)	Attitude, Subjective Norm, Perceived behavioral control
Investments in tracking and tracing systems in the food industry: A PLS analysis	Heyder et al. (2012)	Perceived Usefulness of Tracking and Tracing Systems, Subjective Norm
The effects of charity reputation on charitable giving	Meijer (2009)	Reputation—financial efficiency, media visibility, accreditation status on charitable giving behavior

Factor Identification

Title	Researcher	Factor
Filling the 20 Per Cent Gap: Francis Fukuyama on Trust and Social Capital	Fukuyama, F. (1995)	Trust: The Social Virtues and the Creation of Prosperity
Planned behavior and social capital: Understanding farmers' behavior toward pressurized irrigation technologies	Castillo et al. (2021)	Network, levels of trust, and membership, attitudes, norms, trust
The Impact of Behavioral Drivers on Adoption of Sustainable Agricultural Practices: The Case of Organic Farming in Turkey	Cakirli Akyüz, N., & Theuvsen, L. (2020)	Attitude, Perceived Usefulness

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